

Simplify R:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Theta_{\gamma\gamma}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Theta_{z\beta}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\gamma\gamma}^A \\ -30 \cdot R_z^4 \cdot \Theta_{z\beta}^B \cdot \Theta_{z\beta}^A - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{l} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{l} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -60 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A - 15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3d)$$

Multiply Out Sum:

$$\begin{aligned}
& + \frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{aligned}
& +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A \\
& - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\
& - 60 \cdot R_z^4 \cdot \Theta_{xz}^B \cdot \Theta_{xz}^A - 60 \cdot R_z^4 \cdot \Theta_{yz}^B \cdot \Theta_{yz}^A \\
& - 60 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A \\
& - 15 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\
& + 3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + 3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\
& + 3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\
& + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A \\
& + 3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A + 3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\
& + 3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A + 6 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\
& + 6 \cdot R_z^4 \cdot \Theta_{xy}^B \cdot \Theta_{xy}^A + 6 \cdot R_z^4 \cdot \Theta_{xz}^B \cdot \Theta_{xz}^A \\
& + 6 \cdot R_z^4 \cdot \Theta_{xy}^B \cdot \Theta_{xy}^A + 6 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\
& + 6 \cdot R_z^4 \cdot \Theta_{yz}^B \cdot \Theta_{yz}^A + 6 \cdot R_z^4 \cdot \Theta_{xz}^B \cdot \Theta_{xz}^A \\
& + 6 \cdot R_z^4 \cdot \Theta_{yz}^B \cdot \Theta_{yz}^A + 6 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A
\end{aligned} \right] \quad (4a)
\end{aligned}$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{aligned} &+105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zzz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A \\ &-15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ &-60 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A \\ &-15 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ &+3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + 3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ &+3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\ &+3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A \\ &+3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A + 3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\ &+3 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A + 6 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ &+6 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A + 6 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \end{aligned} \right] \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +24 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 12 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A \\ -12 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A - 12 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A \\ -12 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A + 9 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +3 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A + 3 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +9 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (4c)$$

Combining everything:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \begin{bmatrix} +\frac{8}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - \frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A \\ -\frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A - \frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{zz}^A \\ -\frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{zz}^A + 1 \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +\frac{1}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A + \frac{1}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +1 \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (5)$$

A:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \begin{bmatrix} -\frac{8}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A - \frac{8}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\ -\frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A - \frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\ +\frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A + \frac{4}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +1 \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{1}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{1}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A + 1 \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (6a)$$

$$\xrightarrow{\text{added up}} \begin{bmatrix} -4 \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A - 4 \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\ +\frac{7}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{5}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{5}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A + \frac{7}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (6b)$$

B:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \begin{bmatrix} +4 \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + 4 \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\ +4 \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A + 4 \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +\frac{7}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{5}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{5}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A + \frac{7}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (7a)$$

$$\xrightarrow{\text{added up}} \begin{bmatrix} +\frac{19}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\ +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{19}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (7b)$$